

YOLOWeeds: A novel benchmark of YOLO object detectors for multi-class weed detection in cotton production systems

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ABSTRACT

Weeds are among the major threats to cotton production. Overreliance on herbicides for weed control has accelerated the evolution of herbicide-resistance in weeds and caused increasing concerns about environments, food safety and human health. Machine vision systems for automated/robotic weeding have received growing interest towards the realization of integrated, sustainable weed management. However, in the presence of unstructured field environments and significant biological variability of weeds, it remains a serious challenge to develop reliable weed identification and detection systems. A promising solution to address this challenge are the development of large-scale, annotated image datasets of weeds specific to cropping systems and data-driven AI (artificial intelligence) models for weed detection. Among various deep learning architectures, a diversity of YOLO (You Only Look Once) detectors is well-suited for real-time application and has enjoyed great popularity for generic object detection. This study presents a new dataset (CottoWeedDet12) of weeds important to cotton production in the southern United States (U.S.); it consists of 5648 images of 12 weed classes with a total of 9370 bounding box annotations, collected under natural light conditions and at varied weed growth stages in cotton fields. A novel, comprehensive benchmark of 25 state-of-the-art YOLO object detectors of seven versions including YOLOv3, YOLOv4, Scaled-YOLOv4, YOLOR and YOLOv5, YOLOv6 and YOLOv7, has been established for weed detection on the dataset. Evaluated through the Monte-Carlo cross validation with 5 replications, the detection accuracy in terms of mAP@0.5 ranged from 88.14 % by YOLOv3-tiny to 95.22 % by YOLOv4, and the accuracy in terms of mAP@[0.5:0.95] ranged from 68.18 % by YOLOv3-tiny to 89.72 % by Scaled-YOLOv4. All the YOLO models especially YOLOv5n and YOLOv5s have shown great potential for real-time weed detection, and data augmentation could increase weed detection accuracy. Both the weed detection dataset² and software program codes for model benchmarking in this study are publicly available³, which will be to be valuable resources for promoting future research on big data and AI-empowered weed detection and control for cotton and potentially other crops.

1. Introduction

Cotton [*Gossypium hirsutum* (L.)] is an economically important row crop in the U.S. In 2021, the U.S. harvested 10.0 million acres of cotton (corresponding to more than 18 million 480-pound bales) (USDA-NASS, 2022a), with farmgate value estimated 7.5 billion dollars (USDA-NASS, 2022b). Weeds are threatening cotton production by competing with

crops for resources (e.g., water, nutrients) and providing hosts for diseases and pests. Poor weed management can cause yield loss of more than 50 % (MacRae et al., 2013; Morgan et al., 2001), or even up to 90 % if weeds are left uncontrolled (Manalil et al., 2017).

Since the commercial introduction of *trans*-genetic, herbicide-resistant (HR) crops in 1996, about 96 % of the U.S. acreage for cotton was planted with HR seeds by 2020 (USDA-ERS, 2020). Weed

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³ <https://github.com/DongChen06/DCW>

control of cotton has now become predominantly reliant on broadcast application of herbicides (e.g., glyphosate and dicamba). This, however, has exerted heavy selection pressure for herbicide resistance and led to the emergence of many HR weeds (Kniss, 2018), most notably, glyphosate-resistant weeds such as Palmer Amaranth (Sosnoskie and Culpepper, 2014) and Common Waterhemp (Legleiter and Bradley, 2008), which substantially increase management costs (Gould et al., 2018). Overreliance on repeated application of herbicides has also spurred concerns about the environment, food safety and human health. The U.S. cotton industry, therefore, needs to improve weed control strategies to create a more sustainable weed management system.

Machine vision technologies play a crucial role in future sustainable, integrated weed management based on a diversity of control tactics (Young, 2018). Using imaging sensors and computer algorithms, machine vision systems are able to distinguish weeds from crops, identify weed species and deliver precision and plant-specific weed control. There is a growing interest in developing machine vision-based robotic systems for within-season weed control for crops such as cotton (Barnes et al., 2021; Pandey et al., 2021). Once weeds are detected and localized, appropriate treatments can be applied to remove the weeds, e.g., by spot spraying of herbicides (Lamm et al., 2002; Chostner, 2017), actuating mechanical cultivators/fingers (Kennedy et al., 2020; Van Der Weide et al., 2008), or through thermal methods (Bauer et al., 2020). All these weeding methods have the potential to significantly reduce herbicide consumption, which is necessary for mitigating herbicide resistance in weeds.

Accurate weed identification and detection is crucial for the success of machine vision-based precision weed removal while causing minimal damage to crops. There are some intelligent weeders commercially available (Peruzzi et al., 2017), such as Robovator, by detecting crop row patterns. However, these weeders do not differentiate crops and weeds for precision treatments, and thus are not well suited for intra-row, in-crop weeding. Distinguishing crops from weeds and recognizing weed species is a chronically challenging task (Slaughter et al., 2008). This is largely attributed to significant biological variability of weeds/crops and uncontrolled field environments. Today's state-of-the-art machine vision systems still cannot reliably achieve weed recognition rates well above 95 % (Westwood et al., 2018; Lati et al., 2021), which hinders the practical adoption of machine vision weeding systems. At the time of writing, John Deere (Moline, IL) has introduced machine vision-based spraying technology, See & Spray™ Ultimate, for row crops including cotton, which enables detecting weeds among crops and spraying herbicides to targeted weeds. Currently, the field performance of the technology remains to be fully validated before it can be scaled up and widely adopted.

Two major factors greatly affect weed recognition performance, including the size and quality of image data for model training and weed recognition algorithms. Large-scale labeled image data is key to ensuring the performance of computer vision algorithms. It is reported that the performance of advanced deep learning methods on vision tasks increases logarithmically with the volume of training data (Sun et al., 2017). In precision agriculture, the lack of large-scale, high-quality annotated datasets remains a crucial bottleneck to the full exploitation of deep learning algorithms (Lu and Young, 2020), towards developing robust machine vision systems. For weed recognition, good datasets should provide adequate representations of relevant weed species, environmental conditions (e.g., light conditions, soil types) and growth-stage-related morphological or physiological variations. In addition to demanding domain expertise in weed identification, preparing such datasets is notoriously time-consuming and costly. Some recent studies have been carried out to develop image datasets for weed control (Lu and Young, 2020), e.g., DeepWeeds (Olsen et al., 2019), Early crop weed dataset (Espejo-Garcia et al., 2020), Hedge bindweed (Gao et al., 2020), CottonWeedID15 (Chen et al., 2022), the Eden Library (Mylonas et al., 2022) and Weed-AI (<https://weed-ai.sydney.edu.au/>). So far, to the best of our knowledge, only CottonWeedID15 (Chen et al., 2022) is dedicated

to the recognition of weeds specific to cotton production systems. However, this dataset contains only image-level annotations and thus is not readily suitable for weed detection tasks that require bounding box annotations for weed instances in the images.

Meanwhile, considerable research has been done on image processing/analysis techniques for weed detection (Wang et al., 2019; Meyer and Neto, 2008; Wu et al., 2021). A variety of color indices that accentuate plant greenness have been proposed for enhanced discrimination and segmentation of weeds from soil backgrounds (Meyer and Neto, 2008; Woebbecke et al., 1995). Most of them, despite the ease of computation, are not robust to various imaging conditions, especially in dealing with images acquired under variable natural field light conditions (Hamuda et al., 2016; Bawden et al., 2017; Lu et al., 2022). Recently, data-driven methods based deep learning (DL) algorithms, particularly convolutional neural networks (CNNs), have been used for weed detection (Hasan et al., 2021). Powered by large-scale datasets, well-trained DL models can be robust to biological variability and imaging conditions while achieving satisfactory classification or detection accuracies (Chen et al., 2022; Olsen et al., 2019; Suh et al., 2018). The trained DL models can be deployed on dedicated computing hardware (e.g., NVIDIA Jetson AGX Xavier modules) that fits in a robotic platform towards real-time weed recognition (Du et al., 2021; Olsen et al., 2019).

The weed recognition tasks, fall in three basic categories: 1) classifying images as crops/weeds, 2) detect or localize weed plants in the images, and 3) segment images into semantic weed maps, corresponding to three fundamental problems in computer vision, i.e., image classification, object detection, and semantic segmentation, respectively. DL methods are most commonly used to train weed classification models (Olsen et al., 2019; Espejo-Garcia et al., 2020; Chen et al., 2022; Dyrmann et al., 2016; Suh et al., 2018), given image-level labelled datasets. However, the resultant models provide no information regarding the locations of specific weeds in an image and thus are suboptimal for delivering precision weed control. In comparison, object detection that requires localizing objects of interest within an image (Girshick et al., 2015), would predict the locations of weeds in images and be more beneficial for high-precision weed removal.

There are mainly two types of DL object detectors (Liu et al., 2020), i.e., two-stage detectors that includes a preprocessing step to generate object proposals in the first stage, followed by object classification and bounding box regression in the second stage, and single/one-stage detectors that are end-to-end without involving the process of region proposals. Compared to two-stage detectors such as R-CNN (Girshick et al., 2015), Faster-RCNN (Ren et al., 2015) and Mask-RCNN (He et al., 2017), one-stage detectors are more computationally efficient, faster in inference, and thus more suitable for real-time application, especially for resource-constrained embedded devices. The most notable example of one-stage detectors is presumably You Only Look Once (YOLO), which was originally proposed by Redmon et al. (2016) and subsequently evolved into YOLO9000 (Redmon and Farhadi, 2017) and YOLOv3 (Redmon and Farhadi, 2018). YOLOv3 achieves the best trade-off between accuracy and speed and has become one of the most widely used object detectors. Following YOLOv3, architectural modifications have been made for improved accuracy and/or speed, resulting in versions including YOLOv4 (Bochkovskiy et al., 2020), Scaled-YOLOv4 (Wang et al., 2021a), YOLOv5 (Jocher et al., 2020), and YOLOR (Wang et al., 2021b), YOLOv6 (Li et al., 2022a), and YOLOv7 (Wang et al., 2022), and the newest YOLOv8 (<https://ultralytics.com/>) launched in January of 2023. Each of these YOLO-derived object detectors can be configured with varied levels of model complexity, leading to different implementation variants.

YOLO object detectors have found multiple applications in weed detection (Gao et al., 2020; Sharpe et al., 2020; Ahmad et al., 2021). Gao et al. (2020) reported on YOLOv3-tiny (a simplified version) for detecting hedge bindweed in sugar beet fields and obtained mean average precision (mAP) of 0.829. In Sharpe et al. (2020) on goosegrass detection, a YOLOv3-tiny model achieved F-scores of 0.85 and 0.65 in

strawberry and tomato fields, respectively. Ahmad et al. (2021) used YOLOv3 to detect four weed species commonly in the Midwestern U.S., obtaining an overall mAP of 54.3 %. In evaluating eight object detectors for weed detection, Li et al. (2022b) obtained the best mAP of 79.6 % on a public dataset (Sudars et al., 2020) by YOLOX (Ge et al., 2021). These studies show the efficacy of YOLO detectors, but they only examined a single weed class, except Ahmad et al. (2021) who experimented four-class weeds but with a small dataset. The overall accuracy in Ahmad et al. (2021) was still unsatisfactory, largely because of the limited dataset size. Currently no studies have been conducted on detecting multi-class weeds that are important to U.S. cotton cropping systems. Given active developments in YOLO detectors, it would be greatly beneficial to the research community to establish a comprehensive benchmark of YOLO detectors on multi-class weed detection.

Extended from our previous study on weed classification (Chen et al., 2022), this research is therefore to make the first comprehensive performance evaluation of the state-of-the-art YOLO object detectors for identifying multi-class weeds in U.S. cotton production. Both the weed dataset and software programs developed in this study are made publicly accessible. This study is expected to positively impact future research into developing machine vision-based weeding systems for cotton and potentially other crops. The major contributions of this paper are highlighted as follows:

1. A diverse weed dataset consisting of 5648 images of 12 weed classes with 9370 bounding boxes of weeds important to cotton in the southern U.S.;
2. A comprehensive evaluation and benchmark of 25 state-of-the-art YOLO object detectors was established for multi-class weed detection;
3. The effect of data augmentation on the performance of YOLO models was examined.

2. Materials and methods

2.1. Cotton weed dataset

The weed dataset, CottonWeedDet12 (Lu, 2023), in this study was collected using different hand-held smartphones and digital color cameras (with resolution of more than 10 megapixels) from cotton fields of Mississippi State University research farms, including the R. R. Foil Plant Science Research Center (Starkville, MS), W.B. Andrews Agricultural Systems Research Farm (Starkville, MS) and Black Belt Experiment Station (Brooksville, MS). To ensure diversity and variations of images for model robustness (Lu and Young, 2020), images were collected under natural field light conditions and different weather conditions (e. g., sunny, cloudy, and overcast) at varying stages of weed growth and different field sites. Regular visits to cotton fields, after crop emergence until canopy closure, were conducted throughout June to September of

2021 for weed image collection.

The acquired images were labeled by trained personnel who draw bounding boxes for weed instances in images using the VGG Image Annotator (VIA) (Dutta and Zisserman, 2019). The weed instances were labeled by their common names instead of scientific names. Some weed species that fall in the same plant genus or family and are similar in weed management were assigned with the same weed class label. For instance, all the weed plants of “Ivy Morningglory”, “Pitted Morningglory”, “Entireleaf Morningglory” and “Tall Morningglory” were labeled with the name “Morningglory”. The resultant annotations, which were saved in JavaScript Object Notation (JSON) file format, were visualized and double-checked by weed experts to ensure annotation quality. This eventually led to a dataset with 5960 weed images of 21 weed classes with a total of 9898 bounding box annotations. Because of the imbalance distribution of weed data, as shown in Fig. 1, only the top 12 weed classes (containing more than 140 bounding boxes per class) were considered in this study, resulting in the dataset of CottonWeedDet12 comprising 5648 images with 9370 bounding box annotations, publicly available in the Zenodo repository². This dataset will be updated with more weed images collected and labeled for future growth seasons.

The CottonWeedDet12 dataset, in terms of image number, is more than 10 times larger than the dataset for four weed species presented in (Ahmad et al., 2021), and to our knowledge, it is the first largest public dataset for weed detection pertaining to cotton production systems. Fig. 2 shows example annotated images where only one weed class is highlighted in the images per row. Variations exist between and within weed classes regarding plant morphology/color, soil background and field light conditions, which are desirable for building models robust to imaging conditions. Some images contain weed plants of multiple species. A more detailed analysis of dataset characteristics will be presented in a future study.

2.2. Weed detection

2.2.1. YOLO object detectors

DL-based object detectors are usually composed of two main parts, i. e., a backbone and a head (Bochkovskiy et al., 2020). The backbone extracts features from high-dimensional inputs and is often pretrained on ImageNet data, while the head is used to predict classes and bounding boxes of objects. As to the head, object detectors can be categorized into two-stage and one-stage, depending on if a separate module is used to generate region proposals (Liu et al., 2020). YOLO object detectors all use a one-stage head that classifies and localizes semantic objects in a single shot using dense sampling. Between the backbone and head are some layers often inserted to collect feature mAPs from different stages, which is also known as the neck of an object detector. The advancements in object detection have been focusing on improved backbones and feature integration models as well as various training techniques (e.g., data augmentation, hyper-parameter optimization, batch

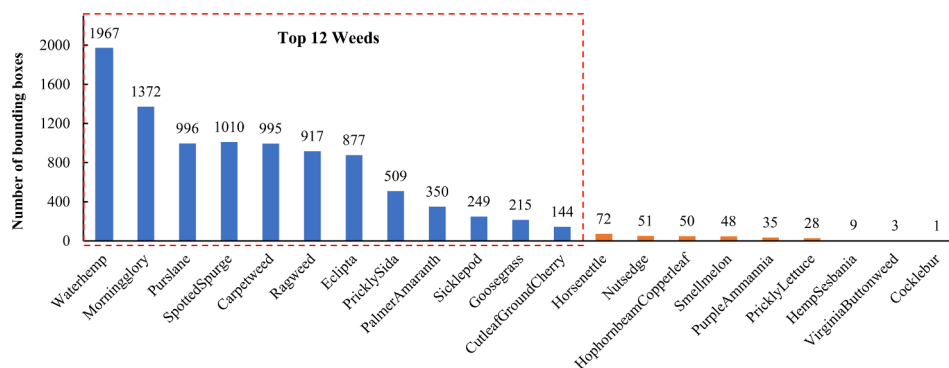


Fig. 1. Bar plot of the cotton weed detection dataset consisting of 5960 images of 21 weed classes with 9898 bounding boxes. Because of weed class imbalance, only the top 12 weed classes with 9370 bounding boxes are used for weed detection in this study.

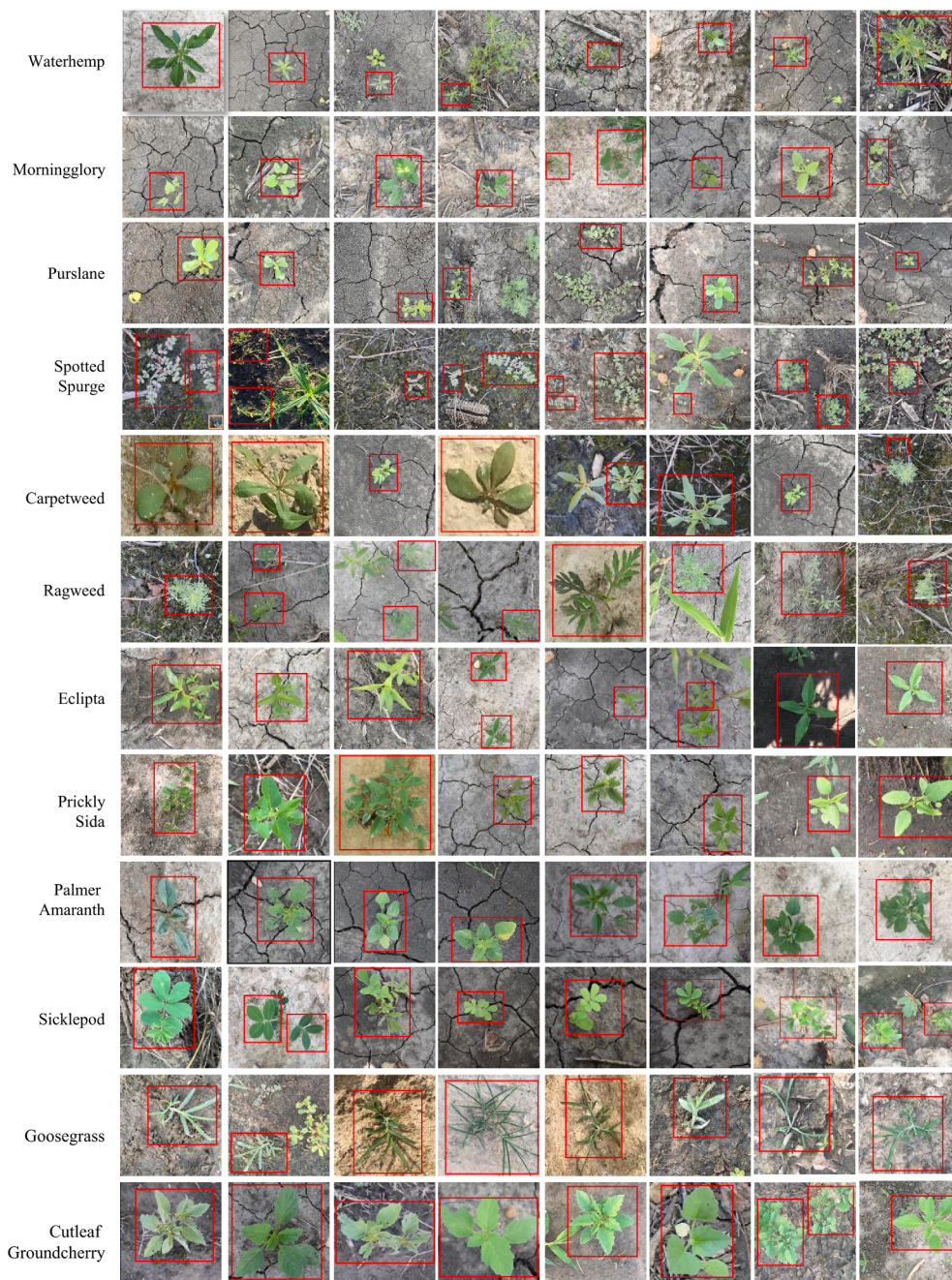


Fig. 2. Example images from the CottonWeedDet12 dataset. Each row displays randomly selected images from each of the top 12 weed classes (see also Fig. 1), with the corresponding weed instances indicated by bounding boxes.

normalization).

YOLOv3 (Redmon and Farhadi, 2018) realizes the optimal trade-off between speed and accuracy for real-time object detection, compared to its two predecessors (Redmon et al., 2016; Redmon and Farhadi, 2017). YOLOv3 uses the Darknet-53 backbone for feature extraction from input images, and its simplified version, YOLOv3-tiny is based on the Darknet-19. **YOLOv4** (Bochkovskiy et al., 2020) differs from YOLOv3 in a number of aspects. It uses the CSPDarknet-53 backbone (Wang et al., 2020), and SPP (spatial pyramid pooling) and PAN (path aggregation network) block neck. To enhance detection accuracy, YOLOv4 incorporates a set of training methods, including “bag of freebies” (methods that only increase training time) such as CutMix (Yun et al., 2019), Mosaic data augmentation (Bochkovskiy et al., 2020) and DropBlock regularization (Ghiasi et al., 2018), and “bag of specials”

(methods that only slightly increase inference time but can significantly increase detection accuracy) such as mish activation (Misra, 2019), cross-stage partial connections and Diou-NMS (Zheng et al., 2020). **Scaled-YOLOv4** (Wang et al., 2021a) scales YOLOv4 for deployment on various types of computing devices (e.g., general, low-end, and high-end GPUs), based on the cross-stage partial network (Wang et al., 2020) for model scaling. Among the scaled models, YOLOv4-large scaled for cloud GPUs achieved the state-of-the-art detection accuracy on the COCO dataset (Wang et al., 2021a). **YOLOR** (Wang et al., 2021b) is one of the latest YOLO object detectors, in addition to its peers like YOLOX (Ge et al., 2021), YOLOF (Chen et al., 2021) and PPYOLOv2 (Huang et al., 2021). Based on the YOLOv4-CSP (Wang et al., 2021a), YOLOR integrates implicit and explicit knowledge to learn a general, unified representation for multiple tasking (e.g., joint object detection and

multi-label image classification). Tested on object detection tasks, YOLOR achieved comparable accuracy as Scaled-YOLOv4 but a significant increase in inference speed (Wang et al., 2021b).

Shortly after the launch of YOLOv4 (Bochkovskiy et al., 2020), YOLOv5 was released by Ultralytics LLC (<https://ultralytics.com/>), which claimed to outperform all its YOLO predecessors (Thuan, 2021). Unlike other YOLO detectors, YOLOv5 was only released in a GitHub repository but not a formal peer-reviewed publication, which, alongside its structural similarity with YOLOv4, has led to some controversy about the validity of their proposed name in the computer vision community. It seems that the only significant modification of YOLOv5 over YOLOv4 was the integration of the anchor box selection process into the model (Nelson and Solawetz, 2020). Nonetheless, YOLOv5 has gained significant popularity as fast and high-performing object detectors and showed better detection accuracy in certain circumstances (Thuan, 2021). The implementation of YOLOv5, in the framework of PyTorch, is actively evolving with the latest version of YOLOv5-v6.1 at the time of writing, and it has flexible control of model size and can be run on a range of devices.

The two very recent YOLO versions are YOLOv6 (Li et al., 2022a) and YOLOv7 (Wang et al., 2022), which both have shown superior performance in real-time object detection. YOLOv6 is featured with a set of renovated designs in the network architecture, label assignment, loss function, data augmentation, and quantization for industrial deployment. YOLOv7, proposed by the authors of YOLOv4 (Bochkovskiy et al., 2020) and Scaled-YOLOv4 (Wang et al., 2021a), implements an extended efficient layer aggregation network and various trainable bag-of-freebies methods (e.g., planned re-parameterization and coarse-to-fine lead guided label assignment) to enhance network training and detection accuracy without increasing inference costs. Trained on the COCO dataset from scratch, YOLOv7 surpassed previous object detectors including YOLOR, YOLOX, Scaled-YOLOv4, YOLOv5 and among others, in both speed and accuracy (Wang et al., 2022).

In this study, the seven types of YOLO object detectors described above, including YOLOv3, YOLOv4, Scaled-YOLOv4, YOLOR, YOLOv5, YOLOv6, and YOLOv7, were selected for developing weed detection models for the cotton weed dataset. Although previous research experimented with some of these detectors for weed detection (Ahmad et al., 2021; Gao et al., 2020; Sharpe et al., 2020; Li et al., 2022b), they only assessed one or two YOLO models for one or a very small number of weed classes. No research has been done on a comprehensive evaluation of a range of YOLO detectors for detecting multi-class weeds relevant to cotton production. It is important to point out that the selected YOLO detectors have open-sourced software packages developed by their creators as summarized in Table 1, which were adapted in this study to train weed detection models as described below.

Table 1
Summary of YOLO variants with open-source software packages.

Index	YOLO models	URL	Reference
1	YOLOv3	https://github.com/ultralytics/YOLOv3	Redmon and Farhadi (2018)
2	YOLOv4	https://github.com/WongKinYiu/PyTorch-YOLOv4	Bochkovskiy et al. (2020)
3	Scaled-YOLOv4	https://github.com/WongKinYiu/ScaledYOLOv4	Wang et al. (2021a)
4	YOLOR	https://github.com/WongKinYiu/yolor	Wang et al. (2021c)
5	YOLOv5	https://github.com/ultralytics/YOLOv5	Jocher et al. (2020)
6	YOLOv6	https://github.com/meituan/YOLOv6	Li et al. (2022a)
7	YOLOv7	https://github.com/WongKinYiu/yolov7	Wang et al. (2022)

2.2.2. Experimentation

Fig. 3 shows the pipeline of modeling for weed detection. The original image annotations in JSON format from VIA (Dutta and Zisserman, 2019) are converted to YOLO format labels for YOLOv3, YOLOv5, YOLOR, Scaled-YOLOv4, YOLOv6, and YOLOv7, and COCO data format for YOLOv4. After the format conversion, the image dataset was randomly partitioned into train, validation, and test subsets according to the partition ratio of 65 % - 20 % - 15 % (3670–1130–848 images), respectively.

To facilitate model training, all the YOLO object detectors were trained via transfer learning (Zhuang et al., 2020) by fine tuning pre-trained weights learnt on the COCO dataset (Lin et al., 2014). The original images were resized to the spatial resolution of 640×640 pixels as training inputs, as dictated by the YOLO network architectures. All the YOLO models were trained with a batch size of 16 for 300 epochs, in the framework of PyTorch (version 1.10) (Paszke et al., 2019). We trained Scaled-YOLOv4 across two GPUs and other YOLO detectors on a single GPU. In YOLOv4, the learning rate was dynamically updated by a polynomial decay learning rate scheduling strategy with an initial learning rate 0.1 and the warm-up steps set to 1000. The cosine annealing method (He et al., 2019) was adopted in all other YOLO detectors to adjust the learning rate over time. All other hyperparameters used the default settings in the official implementations (Table 1). Model training and testing experiments were performed on a Ubuntu 20.04 server with an AMD 7502 32-core processor (128 GB memory) and eight Geforce RTX 3090Ti GPUs (24 GB GDDR6X memory).

Data augmentation can improve model performance in object detection (Zoph et al., 2020), as commonly observed in image classification. Although advanced object detectors like YOLOv4 have incorporated data augmentation in model training, data augmentation independent of object detectors may still lead to performance gains. Hence, to improve detection accuracy, the training data set was augmented by randomly performing one of the ten geometric and photometric transformations illustrated in Fig. 4, to individual training images. Here to save computation costs, only two scenarios of data augmentation were considered, i.e., augmenting the original training set by twice and four times respectively (Fig. 3). These transformations were performed with the aid of the image augmentation package Albumentation (Buslaev et al., 2020). It should be noted that the information of bounding boxes for weeds was preserved during the augmentation. In this study, the data augmentation was only experimented on three YOLO detectors, i.e., YOLOv3 (the worst model without data augmentation in terms of mAP@0.5 as shown in Table 2), YOLOv4 (the best model in terms of mAP@0.5 as shown in Table 2) and YOLOv5, to examine its effectiveness.

Due to the random nature of dataset partition, training detection models for one specific dataset partition is inadequate for objectively evaluating model performance, because the detection results can be heavily dependent upon the particular partition of data. If for some reason, the partition happens to be a good one, the results could be over optimistic. To address the drawback and obtain a reliable estimate of model performance, a Monte-Carlo cross validation (CV) (also known as repeated holdout CV) procedure with replications (Raschka, 2020) was conducted in this study, in which dataset partition and subsequent model training and evaluation were repeated for five times with different random seeds and the mean accuracy on the testing data over the five replications was computed for performance evaluation. Instead of splitting the original data into K equal parts only once in conventional K -fold CV, the employed Monte-Carlo CV procedure partitioned the data randomly into training, validation, and test sets with five replications (Fig. 3) and conceivably would lend more confidence in model performance assessment.

2.3. Performance evaluation metrics

The performance of YOLO object detectors in weed detection was

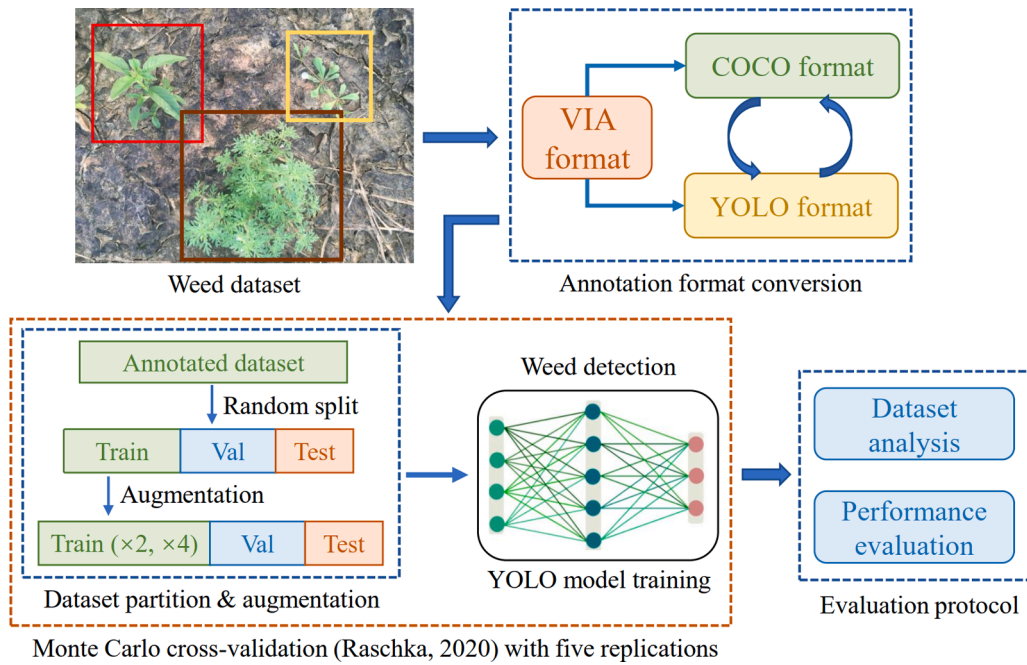


Fig. 3. The proposed pipeline of weed detection by YOLO object detectors.

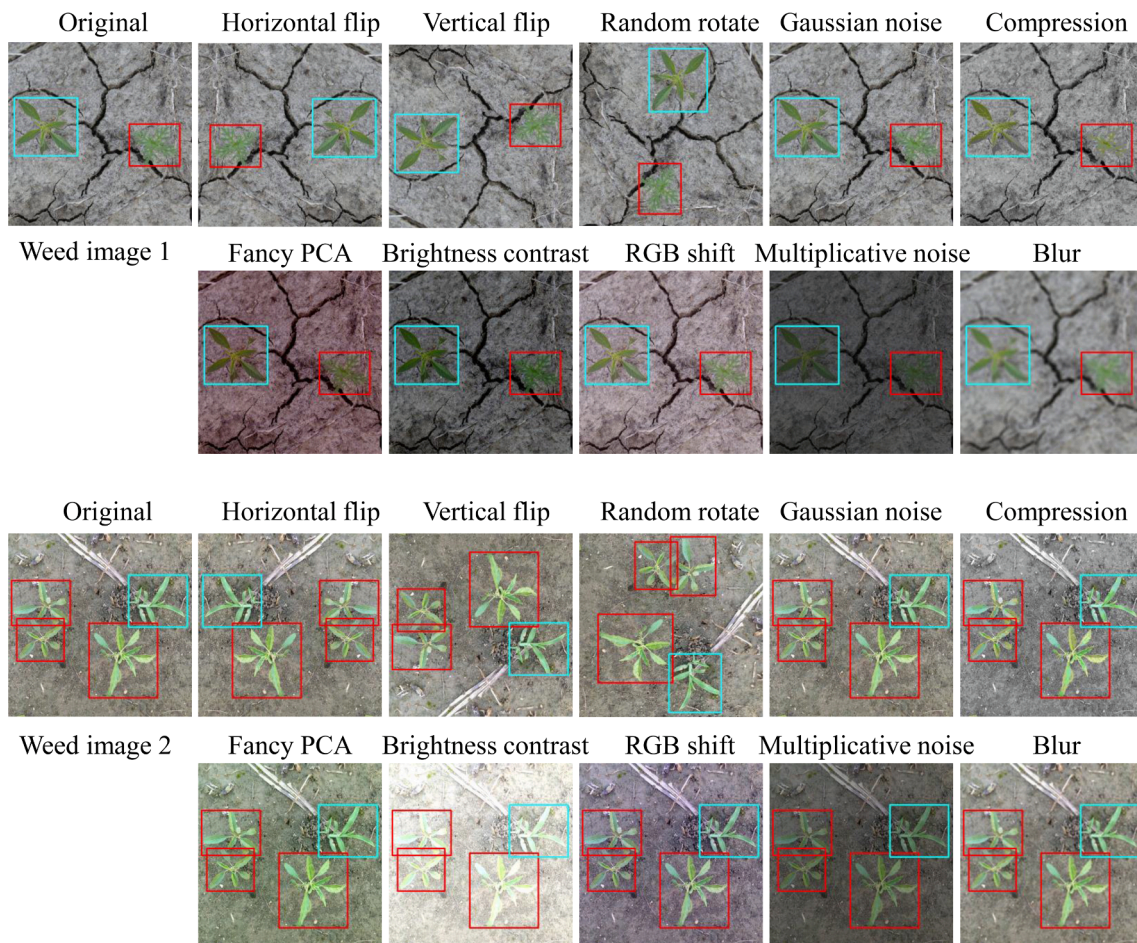


Fig. 4. Illustration of basic image augmentation on two weed images and each image is processed by ten different traditional image augmentation methods. The bounding boxes of difficult colors in the images represent different weed species, and the bounding box information is preserved during image augmentation.

Table 2

Weed detection performance of 25 YOLO detectors on the testing dataset. The accuracies (Precision, Recall and mAP) are in percentage. The mean and standard deviation of test results on five training models are presented for each model. YOLOv3-tiny and YOLOv3 are the versions of YOLOv3 with feature pyramid networks (FPN). YOLOv3-SPP is the version of YOLOv3 with spatial pyramid pooling (SPP) module. YOLOv4-pacsp is the version based on (Jocher et al, 2020).

Index	YOLO models		Precision	Recall	mAP@0.5	mAP@[0.5:0.95]
1	YOLOv3	YOv3-tiny	86.22 ± 1.56	83.32 ± 3.28	88.14 ± 0.63	68.18 ± 1.15
2		YOLOv3	94.00 ± 1.56	90.56 ± 0.94	93.60 ± 0.21	86.98 ± 1.08
3		YOLOv3-SPP	94.82 ± 0.77	89.92 ± 1.09	94.28 ± 0.59	87.50 ± 1.13
		Average	91.68 ± 4.20	87.93 ± 3.89	92.01 ± 2.88	80.89 ± 9.36
4	YOLOv4	YOLOv4pacsp-s	79.18 ± 2.46	94.28 ± 0.64	94.40 ± 0.43	87.04 ± 0.92
5		YOLOv4pacsp	83.36 ± 2.67	94.52 ± 0.85	94.92 ± 0.39	89.26 ± 0.67
6		YOLOv4	81.28 ± 2.55	95.04 ± 0.47	95.22 ± 0.38	89.48 ± 0.26
7		YOLOv4pacsp-x	82.34 ± 2.76	93.18 ± 0.88	93.92 ± 0.65	88.92 ± 0.91
	Scaled-YOLOv4	Average	81.67 ± 2.96	94.26 ± 0.97	94.62 ± 0.67	88.68 ± 1.20
8		YOLOv4-P5	85.40 ± 2.34	94.66 ± 0.39	94.86 ± 0.55	89.44 ± 0.62
9		YOLOv4-P6	85.06 ± 2.03	94.66 ± 0.97	95.08 ± 0.91	89.72 ± 0.87
10		YOLOv4-P7	81.30 ± 2.99	94.96 ± 0.86	95.16 ± 0.82	89.44 ± 0.70
	YOLOR	Average	83.92 ± 3.00	94.76 ± 0.74	95.03 ± 0.73	89.53 ± 0.70
11		YOLOR-P6	83.62 ± 0.34	93.50 ± 0.76	94.36 ± 0.58	89.04 ± 0.89
12		YOLOR-CSP	84.52 ± 1.83	93.84 ± 0.72	94.64 ± 0.48	89.32 ± 0.36
13		YOLOR-CSP-X	81.80 ± 3.19	94.18 ± 0.36	94.36 ± 0.21	89.22 ± 0.72
	YOLOv5	Average	83.31 ± 2.40	93.84 ± 0.66	94.45 ± 0.44	89.19 ± 0.65
14		YOLOv5n	91.98 ± 0.66	88.02 ± 1.06	93.14 ± 0.56	83.64 ± 0.72
15		YOLOv5s	93.86 ± 0.59	89.22 ± 1.27	94.02 ± 0.48	86.60 ± 0.93
16		YOLOv5m	94.78 ± 0.73	89.42 ± 0.93	93.68 ± 0.51	87.10 ± 0.73
17	YOLOv6	YOLOv5l	94.72 ± 1.53	89.50 ± 0.76	93.80 ± 0.67	87.70 ± 0.56
18		YOLOv5x	94.60 ± 0.78	90.12 ± 0.62	94.42 ± 0.62	88.70 ± 0.75
		Average	93.99 ± 1.37	89.26 ± 1.11	93.81 ± 0.67	86.75 ± 1.87
19		YOLOv6n	92.10 ± 0.36	86.17 ± 0.76	93.53 ± 0.93	87.40 ± 0.89
20	YOLOv7	YOLOv6s	93.20 ± 1.21	87.32 ± 1.30	92.80 ± 0.61	87.30 ± 1.11
21		YOLOv6m	93.50 ± 1.68	88.50 ± 1.66	93.03 ± 0.29	88.07 ± 0.32
		Average	92.93 ± 0.74	87.33 ± 1.17	93.12 ± 0.37	87.59 ± 0.42
22		YOLOv7	94.43 ± 1.44	86.93 ± 2.52	92.83 ± 0.71	86.53 ± 1.31
23	YOLOv7D6	YOLOv7x	94.20 ± 1.22	88.17 ± 2.71	93.50 ± 0.53	88.03 ± 0.87
24		YOLOv7E6	92.43 ± 3.98	87.07 ± 3.18	91.37 ± 3.92	85.50 ± 4.16
25		YOLOv7D6	94.47 ± 1.21	88.67 ± 1.72	93.33 ± 1.45	87.33 ± 1.72
		Average	93.88 ± 0.98	87.71 ± 0.85	92.76 ± 0.97	86.85 ± 1.09

evaluated in terms of detection accuracy, model complexity, and inference times (Dang et al., 2022), which are described below.

2.3.1. Detection accuracy

The detection accuracy of trained models was assessed on test data using common metrics for object detection (Everingham et al., 2015; Padilla et al., 2020), including precision (P), recall (R), and mean average precision (mAP, i.e., mAP@0.5 and mAP@[0.5:0.95]). Among the metrics, the mAP is the primary metric in multi-class object detection for evaluating object detectors.

2.3.2. Number of model parameters

The number of model parameters is a direct measure of model complexity and one of the key considerations for real-world deployment of models. Models with a larger number of parameters generally require more memory for deployment. The number of model parameters also affect computation costs and inference times (see Section 2.3.3).

2.3.3. Computation cost and inference time

Floating point operations (FLOPs) are often used to measure the amount of computation (operations) of the model, i.e., how many operations are required to run a single instance of a given model. FLOPs, as another measure of model complexity, were calculated using the THOP library⁴. Inference time, i.e., the time needed for a trained model to make predictions given an input image, is important to real-time application. Here, the inference time for each YOLO detector was calculated as the average time needed to predict all the images of the test set. Unlike FLOPs, inference time depends on computing hardware.

3. Results and discussion

3.1. Performance of YOLO models

Fig. 5 shows the training curves of mAP@0.5 and mAP@[0.5:0.95] for the top 10 detectors in terms of weed detection accuracy. Due to space constraints, the training curves for other models are not shown here. Overall, these models showed promising training performance in terms of fast convergence and high detection accuracies, with 90 % mAP@0.5 and 80 % mAP@[0.5:0.95] attained within 100 training epochs. The training showed that the accuracies leveled off beyond 200 epochs for all the models (including those not displayed in Fig. 5), confirming that training for 300 epochs was sufficient in this study.

Table 2 summarizes the performance results of all the YOLO detection models. The trained models are publicly available at the project GitHub website³. In general, all the 25 models achieved respectable accuracy for detecting 12-class weeds. The mAP@0.5 values ranged from 88.14 % obtained by YOLOv3-tiny to 95.22 % by YOLOv4. In terms of mAP@[0.5:0.95], except YOLOv3-tiny that gave 68.18 % accuracy, all the models achieved the accuracy ranging from 83.64 % to 89.72 %, with 22 out of 25 achieving the accuracy exceeding 86 %. The lower mAP@[0.5:0.95] values than mAP@0.5 is because the former used higher IoU thresholds (implying more stringent criteria) for the AP calculation. Fig. 6 shows the examples of images predicted by YOLOv3-tiny and YOLOv4. Both models yielded visually good predictions for images with diverse and/or cluttered backgrounds, even with densely populated weeds. The predictions on the test images were also compiled into videos for illustration, which are accessible on the GitHub website. Overall, these results demonstrated the efficacy of the selected YOLO object detectors for multi-class weed detection.

To examine the performance difference between the seven types of YOLO detectors, an average accuracy value was calculated by averaging

⁴ <https://github.com/Lyken17/pytorch-OpCounter>

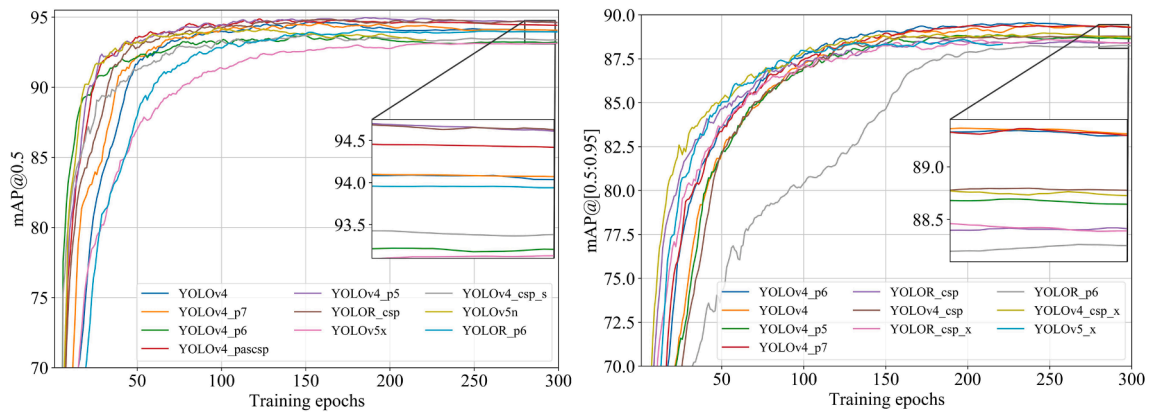


Fig. 5. Training curves of mAP@0.5 and mAP@[0.5:0.95] for the top 10 YOLO models for weed detection.

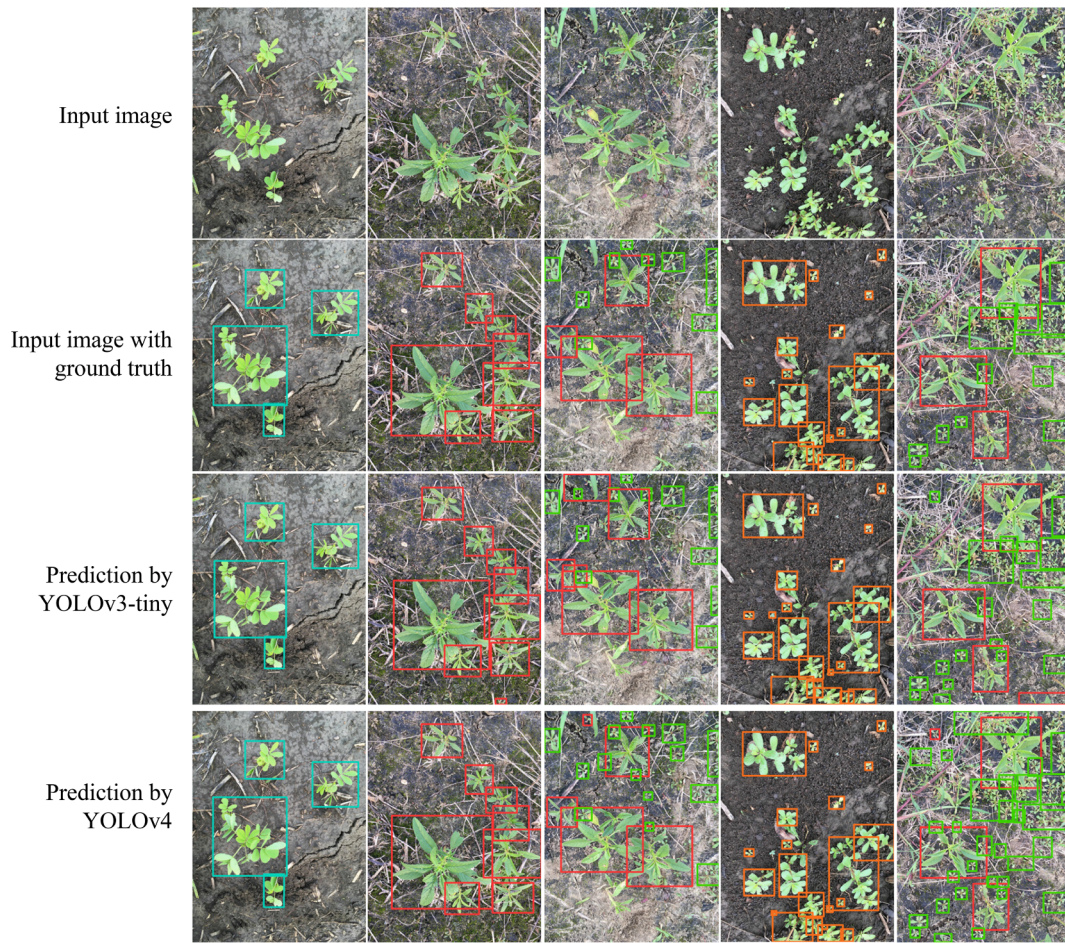


Fig. 6. Examples of weed images with predicted bounding boxes. The bounding boxes with the same color represent the same weed class. The green, red, purple, and orange bounding boxes represent the detection of Purslane, Waterhemp, Spotted Spurge, and Carpetweed, respectively.

the results obtained across different model variants within each detector type (see Table 2). On average, Scaled-YOLOv4 gave the best mAP@0.5 and mAP@[0.5:0.95] of 95.03 % and 89.53 %, respectively, and YOLOv4 and YOLOR achieved high, similar accuracy, followed by YOLOv6, YOLOv7, YOLOv5 and YOLOv3. Although the averaged accuracy may not be a rigorous metric for comparing the different types of YOLO detectors, their difference appeared to suggest that YOLOv4-based detection models (the models indexed at 4–13 in Table 2) performed better than those based on YOLOv5 and YOLOv3 in terms of weed detection accuracy. It is interesting to note that the more recent

YOLO detectors such as YOLOv6 and YOLOv7, while they achieve the state-of-the-art accuracy when benchmarked on the COCO dataset, did not surpass their earlier counterparts including YOLOv4, Scaled-YOLOv4, and YOLOR. The exact reasons for the lower accuracies of the two newer models remain to be investigated, but it should be pointed out that the performance of YOLO detectors can be specific to object detection tasks and datasets, as observed in literature (Nepal and Eslamiat, 2022) and also discussed at <https://github.com/AlexeyAB/darknet/issues/5920>. The *ad-hoc* designed network training and data augmentation tricks in recent YOLO detectors may not

lead to improved performance when transferring the YOLO models pretrained and evaluated on the COCO data to datasets in different domains without dedicated domain adaptation efforts.

Further examination of the detection accuracy for individual weed classes was conducted. Readers are referred to our supplemental material for detailed weed-wise detection results by all the YOLO models⁵. Here only the results obtained by YOLOv3 and YOLOv5s are presented because of space constraints, which are summarized in Table 3. Weed detection accuracy can be affected by factors such as bounding box numbers and size, between-class biological variations, and between-class similarity of weeds. The top 4 weed classes in terms of the number of bounding boxes in the dataset (Fig. 1), including Waterhemp, MorningGlory, Purslane and Spotted Spurge, received good detection accuracy of over 90 % mAP@0.5. Although there were nearly 1000 bounding box annotations for Carpetweed, this class was detected with the lowest accuracy for both YOLOv3 and YOLOv5s with the mAP@[0.5:0.95] of 72.6 % and 71.78 %, respectively. This is presumably because Carpetweed plants in images are overall comparably small and young seedlings, which can be difficult to localize accurately. In addition, Goosegrass was also noticeably challenging for both YOLOv3 and YOLOv5 that resulted in mAP@[0.5:0.95] of 79.26 % and 80.00 %, respectively. As the only grassy weed in the dataset, Goosegrass had the second smallest number of annotations (Fig. 1), which may explain its low detection accuracy. Including more representative training samples alongside with optimized training strategies could help improve the detection accuracy.

Model complexity and inference time are important to practical deployment in resource-constrained settings. The first two subplots in Fig. 7 depicts the GFLOPs and inference times versus number of model parameters for all the tested YOLO detectors. The GLOPs and inference times tended to increase linearly with the number of model parameters. Among the seven types of YOLO models, Scaled-YOLOv4 seemed to be the highest GFLOPs with the longest inference time of 21.62 ms. In contrast, YOLOv5 models, especially the two simplified versions YOLOv5n and YOLOv5s, were the most computationally efficient and the fastest in inference (consuming less 2 ms). Fig. 7 also depicts model inference times versus mAP@[0.5:0.9]. There are likely trade-offs in model selection between accuracy and inference times, given that high accuracy tends to be associated with longer inference times. Overall, all the YOLO detectors experimented in this study have the potential to be deployed for real-time weed detection at tens or even hundreds of frames per second. As models were tested on advanced computing hardware, further testing these models on embedded devices has yet to be done, though.

3.2. Data augmentation

To examine the effect of the data augmentation approaches described in Section 2.2.2 on model performance, further experiments were conducted on three selected YOLO detectors, i.e., YOLOv3-tiny (the worst-performant in terms of mAP@0.5), YOLOv4 (the best model on mAP@0.5) and YOLOv5s, with and without data augmentation, using the Monte-Carlo cross validation procedures with three replications. Fig. 8 shows the training curves for the three YOLO detectors. Clearly, data augmentation substantially speeded up the training process for all three models. Both YOLOv3-tiny and YOLOv4 garnered performance gains in terms of mAP@0.5 and mAP@[0.5:0.95]. Table 4 shows the comparison of the YOLO detectors with and without data augmentation regarding detection accuracy metrics. Data augmentation had positive impacts on the performance of YOLOv4-tiny and YOLOv4. Using the data augmented by four times, YOLOv3-tiny achieved a noticeable increase in mAP@[0.5:0.95], from

67.83 % to 70.83 %. YOLOv4 also saw a slight increase in mAP@0.5 and mAP@[0.5:0.95], although advanced data augmentation (e.g., CutMix and Mosaic data augmentation) was inherent in its modeling process. However, mixed performance was observed for YOLOv5s, suggesting that the proposed data augmentation in Section 2.2.2 may not be effective for YOLOv5s. Given the fact that YOLOv5 detectors have incorporated standard data augmentation approaches⁶, further non-tailored data augmentation may not necessarily contribute to accuracy improvements. Further studies are needed on optimized data augmentation strategies for YOLO detectors for enhanced weed detection.

3.3. Discussion

Research on multi-class weed detection/localization has been scant in literature. This is partly due to the lack of suitable weed datasets with multi-class bounding box annotations. As machine vision-based weeding systems are evolving towards next-generation higher-precision and weed-specific controls, differentiation among weed species and detection of individual weed instances are becoming an increasingly important task. While considerable efforts have been made on weed image classification (Espejo-Garcia et al., 2020; Lu and Young, 2020; Olsen et al., 2019), very few studies were focused on the detection of multi-class weeds, especially with large-scale, multi-class weed data. Extending from our previous work on multi-class weed classification (Chen et al., 2022), this study represents a unique contribution to the research community on weed detection and control by creating and releasing a 12-class weed dataset with nearly ten thousand of bounding box annotations collected under natural field conditions. Moreover, a comprehensive benchmark of state-of-the-art YOLO detectors for weed detection was built with impressive performance in terms of high detection accuracy and fast inference times. The detection accuracies are compared favorably with those reported in previous studies on weed detection using YOLOv3 (Ahmad et al., 2021; Sharpe et al., 2020; Gao et al., 2020).

This study has some limitations that have been identified for further improvements. Although the study did not intend to evaluate all YOLO-derived object detectors for weed detection, there are some other high-performant YOLO models that were not experimented, such as YOLOX (Ge et al., 2021), PP-YOLOv2 (Huang et al., 2021) and the latest YOLOv8. These models will be tested and incorporated in our benchmark that will be actively updated³, as the weed detection dataset is expanded with future efforts. The data augmentation experiments of this study need to be improved in a more principled manner. Cross-dataset evaluation is needed for assessing the robustness or generalization capabilities of data augmentation approaches. Key factors that can potentially affect model performance, such as the number of augmented images and image augmentation or synthesis approaches, should be investigated in depth. This can be greatly beneficial for reducing efforts and resources needed for image collection and annotation. In this regard, it is important to point out that generative adversarial networks (GANs) have gained increasing popularity as a novel approach to synthesizing realistic images for deep learning in agriculture (Lu et al., 2022). A dedicated study is ongoing to evaluate whether GANs and other generative modeling techniques can generate high-quality weed images to improve weed detection performance (Chen et al., 2022).

Moreover, the weed detection models developed in this study remain to be systematically validated by deploying them on a machine vision hardware prototype in future growth seasons. Emphasis will be placed on ascertaining the *out-of-distribution* robustness of models (Hendrycks et al., 2021), which is crucial for practical application. Preliminary testing of the YOLOv4 model (that achieved the best mAP@0.5 in this study) for the growth season of 2022 confirmed the efficacy of the model. The corresponding demos on real-time weed detection are

⁵ https://github.com/DongChen06/DCW/blob/main/datasets/class-wise_statistics.pdf

⁶ <https://github.com/ultralytics/yolov5/pull/3882>

Table 3

Class-wise performance of YOLOv3 and YOLOv5s. P, R and mAP represent precision, recall and mean average precision, respectively.

Index	Weed Class	YOLOv3				YOLO-v5s			
		P	R	mAP @0.5	mAP @[0.5:0.95]	P	R	mAP @0.5	mAP @[0.5:0.95]
1	Waterhemp	93.72	91.14	94.86	90.30	94.42	89.28	95.46	90.54
2	Morninglory	98.28	94.00	96.76	90.28	97.22	93.08	96.92	88.62
3	Purslane	95.20	85.76	91.16	82.46	93.70	85.86	92.74	81.16
4	SpottedSpurge	92.38	86.46	91.62	81.50	91.20	85.00	91.36	80.86
5	Carpetweed	81.50	78.62	84.94	72.62	81.92	75.82	84.16	71.78
6	Ragweed	89.36	90.14	94.44	85.62	89.00	87.12	95.06	85.32
7	Eclipta	94.98	94.84	96.60	92.96	94.88	94.74	98.06	94.10
8	PricklySida	96.76	96.10	96.72	93.22	96.26	96.22	97.64	93.40
9	PalmerAmaranth	97.16	90.68	93.96	90.12	98.84	90.18	94.64	89.48
10	Sicklepod	96.78	97.90	97.50	95.34	96.42	97.90	97.20	94.46
11	Goosegrass	95.52	85.82	88.04	79.26	96.56	84.16	89.00	80.00
12	CutleafGroundcherry	96.48	95.26	96.28	90.08	96.02	91.08	95.78	89.46

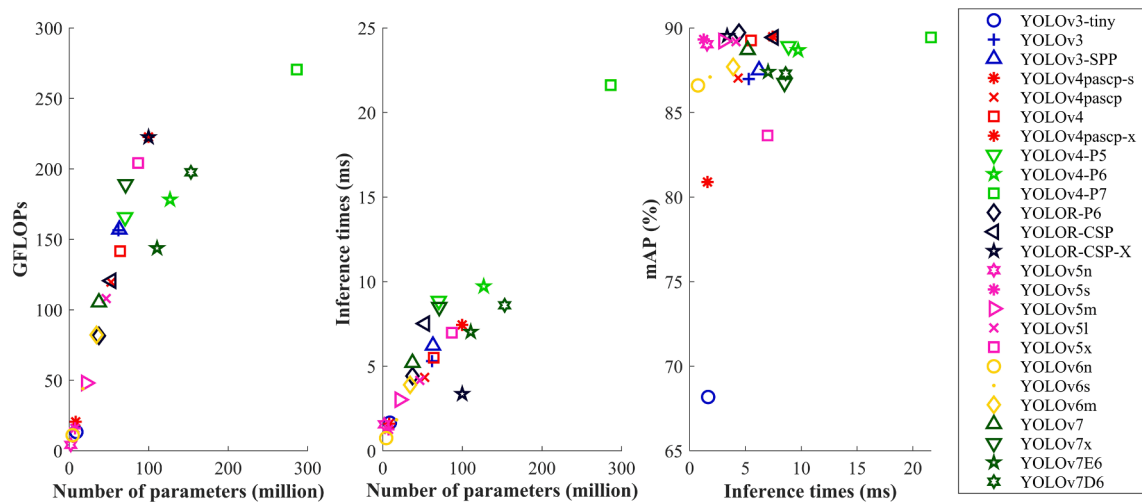


Fig. 7. GFLOPs and inference time (ms) v.s. number of parameters (million), and mAP@[0.5:0.9] versus inference times. The detection models within the same type of YOLO detectors are labeled with markers of the same color. GFLOPs stands for giga float point operations (FPLOPs), which is equal to 10^9 FLOPs, and mAP denotes mean average precision.

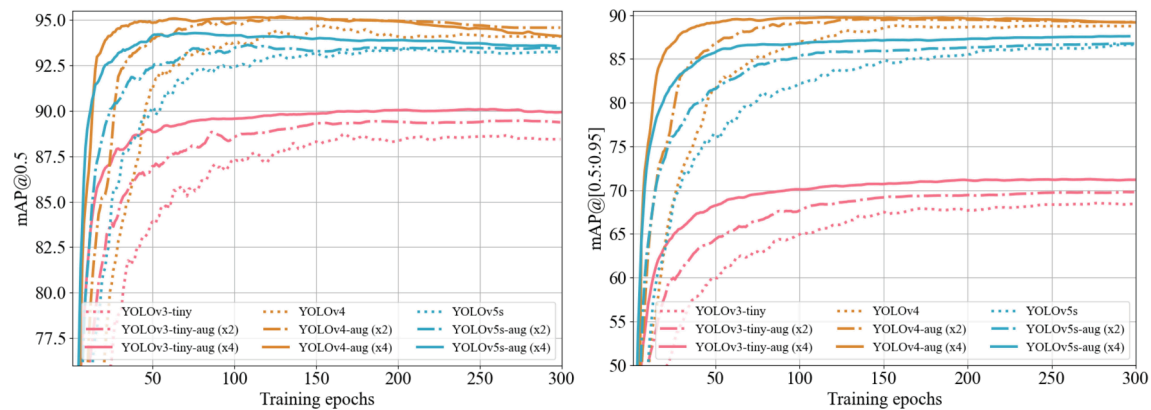


Fig. 8. Training accuracy curves for three YOLO models with and without data augmentation.

publicly available³ for the videos taken from fallow cotton fields at the Plant Science Research Center of Mississippi State University in August of 2022. However more extensive, dedicated research is needed to understand and improve the robustness on real-world distribution shifts encountered under varying imaging conditions.

4. Conclusion

Weed detection is important to precise localization of weeds by machine vision systems for plant-specific weed control. At the core of effective weed detection are the curation of a large-scale, precisely labelled weed dataset and whereby the development of supervised learning models. This paper presents the, to date, the largest weed detection dataset relevant to the U.S. cotton production systems,

Table 4

Weed detection performance of three YOLO models on the cotton weed detection dataset without and with data augmentation. The mean and standard deviation are based on three repeated runs.

Models	Precision	Recall	mAP@0.5	mAP@[0.5:0.95]
YOLOv3-tiny	86.53 ± 0.57	81.67 ± 2.70	88.13 ± 0.81	67.83 ± 1.39
YOLOv3-tiny & Augmentation (×2)	85.43 ± 1.27	84.47 ± 0.61	88.80 ± 0.92	69.47 ± 1.53
YOLOv3-tiny & Augmentation (×4)	86.17 ± 1.91	85.27 ± 1.59	89.80 ± 1.51	70.83 ± 1.76
YOLOv4	82.50 ± 1.91	95.00 ± 0.26	95.30 ± 0.44	89.47 ± 0.31
YOLOv4 & Augmentation (×2)	80.90 ± 2.77	95.47 ± 0.15	95.33 ± 0.58	89.87 ± 0.40
YOLOv4 & Augmentation (×4)	80.57 ± 2.06	96.07 ± 0.06	95.63 ± 0.68	90.00 ± 0.70
YOLOv5s	93.47 ± 0.25	89.27 ± 1.38	94.07 ± 0.67	86.50 ± 1.22
YOLOv5s & Augmentation (×2)	94.10 ± 0.30	88.33 ± 1.07	93.60 ± 0.56	85.97 ± 1.39
YOLOv5s & Augmentation (×4)	93.70 ± 1.31	89.97 ± 0.93	93.77 ± 1.10	86.50 ± 0.79

comprising 5648 images of 12 weed classes with a total of 9370 bounding box annotations, collected under various natural field light conditions. A comprehensive benchmark suite of 25 selected YOLO object detectors was established through transfer learning for multi-class weed detection, which was evaluated in terms of detection accuracy, model complexity and inference times. The YOLO detection models achieved mAP@0.5 scores ranging from 88.14 % by YOLOv3-tiny to 95.22 % by YOLOv4. Overall, the YOLOv4-based models yielded higher accuracy than YOLOv3 and YOLOv5, while YOLOv5n and YOLOv5s showed advantages in fast inference times while achieving comparable detection accuracies. Data augmentation could have positive impacts on the performance of YOLO detectors especially YOLOv3-tiny. This performance benchmark expects to serve as a reference source for informed choices of YOLO object detectors for weed detection and can also be beneficial for object detection tasks in precision agriculture. Both the weed detection dataset and source codes for model development and evaluation are made publicly accessible for the research community.

CRedit authorship contribution statement

Fengying Dang: Formal analysis, Software, Writing – original draft. **Dong Chen:** Formal analysis, Software, Writing – original draft. **Yuzhen Lu:** Conceptualization, Investigation, Data curation, Supervision, Funding acquisition, Project administration, Validation, Writing - original draft, Writing - review & editing. **Zhaojian Li:** Resources, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data set and software programs used in this study are publicly available at <https://doi.org/10.5281/zenodo.7535814> and <https://github.com/DongChen06/DCW>, respectively.

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